

Table 1: Test results for examples 1 – 3.

N	Pdim	SP	MKAS			PBDFF			SMDFF			SMBFGS		
			Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm
1	1000	x1	21	35	0.2140	9.85E-11	44	84	0.1298	1.32E-11	23	75	0.1030	3.21E-11
	1000	x2	3	7	0.0155	0	13	19	0.0258	1.01E-11	18	48	0.0311	2.58E-11
	1000	x3	1	4	0.0709	0	12	21	0.0177	3.29E-11	***	***	***	***
	1000	x4	1	3	0.0045	0	11	18	0.0159	1.56E-11	***	***	***	***
	1000	x5	1	3	0.0057	0	12	17	0.0177	3.84E-11	***	***	***	***
	1000	x6	1	4	0.0047	0	1	13	0.0090	0	1	13	0.0056	0
	5000	x1	23	44	0.2025	7.40E-11	86	145	0.3275	8.12E-12	26	79	0.1267	6.09E-11
	5000	x2	3	7	0.0142	0	13	19	0.0522	1.02E-11	18	48	0.0938	2.28E-11
	5000	x3	1	4	0.0098	0	12	21	0.0660	7.35E-11	***	***	***	***
	5000	x4	1	3	0.0064	0	11	18	0.0442	3.49E-11	***	***	***	***
	5000	x5	1	3	0.0077	0	12	17	0.0524	8.60E-11	***	***	***	***
	5000	x6	1	4	0.0088	0	1	13	0.0264	0	1	13	0.0235	0
	10000	x1	22	42	0.1220	4.95E-11	85	140	0.4786	7.61E-11	25	80	0.2118	4.84E-11
	10000	x2	3	7	0.0191	0	13	19	0.0797	1.02E-11	18	48	0.1389	2.25E-11
	10000	x3	1	4	0.0108	0	13	22	0.1040	5.89E-12	***	***	***	***
	10000	x4	1	3	0.0101	0	11	18	0.0597	4.94E-11	***	***	***	***
	10000	x5	1	3	0.0102	0	13	18	0.0799	6.89E-12	***	***	***	***
	10000	x6	1	4	0.0108	0	1	13	0.0565	0	1	13	0.0277	0
2	1000	x1	1	4	0.0465	0	38	60	0.0369	7.23E-11	40	323	0.1181	4.56E-11
	1000	x2	43	53	0.0321	6.09E-11	49	65	0.0592	8.45E-12	13	110	0.0488	2.94E-11
	1000	x3	1	4	0.0049	0	55	84	0.0449	6.31E-11	***	***	***	***
	1000	x4	1	4	0.0037	0	45	65	0.0487	9.25E-11	***	***	***	***
	1000	x5	1	4	0.0048	0	47	65	0.0374	8.33E-11	29	179	0.0548	9.69E-11
	1000	x6	3	24	0.0088	0	54	89	0.0844	8.08E-11	***	***	***	***
	5000	x1	1	4	0.0106	0	31	71	0.1950	9.05E-11	31	229	0.3156	6.99E-11
	5000	x2	17	24	0.0097	0	28	41	0.1147	7.73E-12	13	110	0.1928	2.92E-11
	5000	x3	1	4	0.0097	0	70	118	0.3157	5.53E-11	***	***	***	***
	5000	x4	1	4	0.0088	0	42	59	0.1595	8.70E-11	***	***	***	***
	5000	x5	1	4	0.0085	0	43	67	0.1861	5.06E-11	***	***	***	***
	5000	x6	3	27	0.0334	0	50	78	0.2275	7.79E-11	***	***	***	***
	10000	x1	1	4	0.0122	0	38	92	0.3153	6.44E-11	30	248	0.5768	1.49E-11
	10000	x2	28	39	0.1486	4.42E-11	41	59	0.2739	3.58E-11	13	110	0.2790	2.92E-11
	10000	x3	1	4	0.0117	0	69	120	0.4739	6.15E-12	***	***	***	***
	10000	x4	1	4	0.0130	0	41	64	0.3267	7.62E-11	***	***	***	***
	10000	x5	1	4	0.0117	0	49	80	0.3601	7.36E-11	***	***	***	***
	10000	x6	3	28	0.0489	0	53	81	0.4004	7.66E-11	***	***	***	***
3	1000	x1	1	3	0.0908	0	11	14	0.0304	2.07E-11	***	***	***	***
	1000	x2	1	3	0.0050	0	10	13	0.0137	9.23E-11	11	22	0.0288	8.27E-25
	1000	x3	1	3	0.0109	0	13	17	0.0110	6.09E-12	10	22	0.0164	4.44E-22
	1000	x4	1	3	0.0159	0	10	13	0.0231	3.86E-11	9	19	0.0211	6.80E-22
	1000	x5	1	3	0.0050	0	11	13	0.0175	5.82E-11	***	***	***	***
	1000	x6	1	3	0.0037	0	11	17	0.0174	1.46E-11	***	***	***	***
	5000	x1	1	3	0.0104	0	11	14	0.0426	4.61E-11	***	***	***	***
	5000	x2	1	3	0.0091	0	10	13	0.0472	9.23E-11	***	***	***	***
	5000	x3	1	3	0.0081	0	13	17	0.0606	1.36E-11	11	24	0.0782	0
	5000	x4	1	3	0.0071	0	10	13	0.0661	8.63E-11	9	19	0.0515	0
	5000	x5	1	3	0.0083	0	12	14	0.0535	7.37E-12	***	***	***	***
	5000	x6	1	3	0.0212	0	11	17	0.0594	3.27E-11	8	20	0.0523	0
	10000	x1	1	3	0.0108	0	11	14	0.0827	6.52E-11	***	***	***	***
	10000	x2	1	3	0.0099	0	10	13	0.0648	9.23E-11	***	***	***	***
	10000	x3	1	3	0.0098	0	13	17	0.1023	1.93E-11	***	***	***	***
	10000	x4	1	3	0.0109	0	11	14	0.0716	6.91E-12	8	17	0.0736	1.40E-19
	10000	x5	1	3	0.0108	0	12	14	0.0929	1.04E-11	***	***	***	***
	10000	x6	1	3	0.0105	0	11	17	0.0823	4.62E-11	8	20	0.0837	0
	1000	x1	1	3	0.0908	0	11	14	0.0304	2.07E-11	***	***	***	***
	1000	x2	1	3	0.0050	0	10	13	0.0137	9.23E-11	11	22	0.0288	8.27E-25
	1000	x3	1	3	0.0109	0	13	17	0.0110	6.09E-12	10	22	0.0164	4.44E-22
	1000	x4	1	3	0.0159	0	10	13	0.0231	3.86E-11	9	19	0.0211	6.80E-22
	1000	x5	1	3	0.0050	0	11	13	0.0175	5.82E-11	***	***	***	***
	1000	x6	1	3	0.0037	0	11	17	0.0174	1.46E-11	***	***	***	***
	5000	x1	1	3	0.0104	0	11	14	0.0426	4.61E-11	***	***	***	***
	5000	x2	1	3	0.0091	0	10	13	0.0472	9.23E-11	***	***	***	***
	5000	x3	1	3	0.0081	0	13	17	0.0606	1.36E-11	11	24	0.0782	0
	5000	x4	1	3	0.0071	0	10	13	0.0661	8.63E-11	9	19	0.0515	0
	5000	x5	1	3	0.0083	0	12	14	0.0535	7.37E-12	***	***	***	***
	5000	x6	1	3	0.0212	0	11	17	0.0594	3.27E-11	8	20	0.0523	0
	10000	x1	1	3	0.0108	0	11	14	0.0827	6.52E-11	***	***	***	***
	10000	x2	1	3	0.0099	0	10	13	0.0648	9.23E-11	***	***	***	***
	10000	x3	1	3	0.0098	0	13	17	0.1023	1.93E-11	***	***	***	***
	10000	x4	1	3	0.0109	0	11	14	0.0716	6.91E-12	8	17	0.0736	1.40E-19
	10000	x5	1	3	0.0108	0	12	14	0.0929	1.04E-11	***	***	***	***
	10000	x6	1	3	0.0105	0	11	17	0.0823	4.62E-11	8	20	0.0837	0

Table 2: Test results for examples 4 – 6.

N	PdIm	SP	MKAS			PBDFF			SMDFF			SMBFGS		
			Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm
4	1000	x1	23	45	0.2055	5.14E-11	30	44	0.0690	8.99E-11	49	148	0.1249	4.25E-11
	1000	x2	27	50	0.0537	4.36E-11	47	55	0.0650	9.98E-11	62	199	0.1436	2.84E-11
	1000	x3	28	44	0.0336	1.41E-11	41	46	0.0575	8.57E-11	49	147	0.1068	9.80E-11
	1000	x4	21	38	0.0263	7.60E-11	53	59	0.0698	9.93E-11	52	159	0.1091	8.01E-11
	1000	x5	21	42	0.0331	5.76E-11	35	39	0.0358	5.19E-11	53	158	0.1093	6.30E-11
	1000	x6	21	35	0.0232	9.24E-11	41	48	0.0585	7.78E-11	38	132	0.0707	5.84E-11
	5000	x1	19	44	0.1102	2.41E-11	33	43	0.2066	9.93E-11	56	170	0.4370	8.78E-11
	5000	x2	23	47	0.1243	9.95E-11	39	45	0.2263	9.88E-11	53	157	0.4390	9.69E-11
	5000	x3	22	47	0.1339	5.41E-11	48	53	0.2651	9.82E-11	45	124	0.3614	5.55E-11
	5000	x4	19	40	0.0987	9.92E-11	38	46	0.2527	4.32E-11	50	146	0.3939	8.10E-11
	5000	x5	23	45	0.1342	2.29E-11	43	47	0.2281	4.41E-11	53	164	0.4045	6.76E-11
	5000	x6	17	34	0.0955	9.62E-11	52	56	0.2820	4.31E-11	28	81	0.2836	7.46E-11
	10000	x1	20	49	0.1843	5.54E-11	35	52	0.3449	8.66E-12	50	147	0.6590	2.56E-11
	10000	x2	24	48	0.5689	6.02E-11	36	42	0.3373	5.45E-11	49	144	0.6555	3.66E-11
	10000	x3	25	50	0.3567	9.25E-11	42	47	0.4547	9.22E-11	45	135	0.6699	9.35E-11
	10000	x4	24	46	0.3521	7.83E-11	36	47	0.3510	5.32E-11	55	172	0.6986	1.54E-11
	10000	x5	24	47	0.3417	9.14E-11	41	47	0.3827	9.02E-11	58	177	0.7569	4.48E-11
	10000	x6	20	44	0.2939	6.08E-11	27	39	0.2583	3.90E-11	33	105	0.4724	2.18E-11
5	1000	x1	1	4	0.1144	0	9	18	0.0226	3.75E-11	10	82	0.0289	1.23E-11
	1000	x2	1	4	0.0110	0	8	17	0.0188	9.65E-11	9	74	0.0327	2.19E-11
	1000	x3	1	4	0.0056	0	11	20	0.0166	6.80E-12	12	100	0.0336	4.50E-12
	1000	x4	1	4	0.0095	0	9	18	0.0167	8.33E-12	9	74	0.0330	5.59E-11
	1000	x5	1	4	0.0071	0	10	18	0.0130	1.45E-11	10	82	0.0252	4.70E-11
	1000	x6	1	4	0.0109	0	9	19	0.0248	7.34E-12	9	75	0.0367	2.21E-11
	5000	x1	1	4	0.0129	0	9	18	0.0435	8.35E-11	10	82	0.1285	2.74E-11
	5000	x2	1	4	0.0099	0	8	17	0.0483	9.65E-11	9	74	0.1073	2.19E-11
	5000	x3	1	4	0.0151	0	11	20	0.0536	1.52E-11	12	100	0.1542	1.01E-11
	5000	x4	1	4	0.0154	0	9	18	0.0488	1.86E-11	10	82	0.1416	5.43E-12
	5000	x5	1	4	0.0139	0	10	18	0.0641	3.24E-11	11	90	0.1505	4.56E-12
	5000	x6	1	4	0.0114	0	9	19	0.0517	1.64E-11	9	75	0.1293	4.94E-11
	10000	x1	1	4	0.0166	0	10	19	0.1055	3.44E-12	10	82	0.1940	3.87E-11
	10000	x2	1	4	0.0224	0	8	17	0.0667	9.65E-11	9	74	0.1649	2.19E-11
	10000	x3	1	4	0.0224	0	11	20	0.0964	2.15E-11	12	100	0.2289	1.42E-11
	10000	x4	1	4	0.0343	0	9	18	0.0609	2.63E-11	10	82	0.1899	7.68E-12
	10000	x5	1	4	0.0171	0	10	18	0.0797	4.58E-11	11	90	0.2225	6.45E-12
	10000	x6	1	4	0.0161	0	9	19	0.0806	2.32E-11	9	75	0.2022	6.99E-11
6	1000	x1	1	5	0.0677	0	1	13	0.0104	0	1	13	0.0109	0
	1000	x2	1	5	0.0101	0	1	13	0.0147	0	1	13	0.0150	0
	1000	x3	26	88	0.0779	6.22E-11	1	13	0.0066	0	1	13	0.0145	0
	1000	x4	1	5	0.0163	0	1	13	0.0068	0	1	13	0.0135	0
	1000	x5	1	5	0.0058	0	1	13	0.0164	0	1	13	0.0149	0
	1000	x6	1	4	0.0067	0	1	13	0.0106	0	1	13	0.0105	0
	5000	x1	1	5	0.0210	0	1	13	0.0212	0	1	13	0.0343	0
	5000	x2	1	5	0.0146	0	1	13	0.0403	0	1	13	0.0265	0
	5000	x3	3	27	0.0621	0	1	13	0.0462	0	1	13	0.0337	0
	5000	x4	1	5	0.0173	0	1	13	0.0358	0	1	13	0.0304	0
	5000	x5	1	5	0.0126	0	1	13	0.0309	0	1	13	0.0450	0
	5000	x6	1	4	0.0158	0	1	13	0.0414	0	1	13	0.0331	0
	10000	x1	1	5	0.0318	0	1	13	0.0404	0	1	13	0.0470	0
	10000	x2	1	5	0.0286	0	1	13	0.0482	0	1	13	0.0356	0
	10000	x3	3	28	0.0867	0	1	13	0.0504	0	1	13	0.0536	0
	10000	x4	1	5	0.0231	0	1	13	0.0498	0	1	13	0.0410	0
	10000	x5	1	5	0.0224	0	1	13	0.0493	0	1	13	0.0463	0
	10000	x6	1	4	0.0233	0	1	13	0.0364	0	1	13	0.0608	0
	1000	x1	1	5	0.0677	0	1	13	0.0104	0	1	13	0.0109	0
	1000	x2	1	5	0.0101	0	1	13	0.0147	0	1	13	0.0150	0
	1000	x3	26	88	0.0779	6.22E-11	1	13	0.0066	0	1	13	0.0145	0
	1000	x4	1	5	0.0163	0	1	13	0.0068	0	1	13	0.0135	0
	1000	x5	1	5	0.0058	0	1	13	0.0164	0	1	13	0.0149	0
	1000	x6	1	4	0.0067	0	1	13	0.0106	0	1	13	0.0105	0
	5000	x1	1	5	0.0210	0	1	13	0.0212	0	1	13	0.0343	0
	5000	x2	1	5	0.0146	0	1	13	0.0403	0	1	13	0.0265	0
	5000	x3	3	27	0.0621	0	1	13	0.0462	0	1	13	0.0337	0
	5000	x4	1	5	0.0173	0	1	13	0.0358	0	1	13	0.0304	0
	5000	x5	1	5	0.0126	0	1	13	0.0309	0	1	13	0.0450	0
	5000	x6	1	4	0.0158	0	1	13	0.0414	0	1	13	0.0331	0
	10000	x1	1	5	0.0318	0	1	13	0.0404	0	1	13	0.0470	0
	10000	x2	1	5	0.0286	0	1	13	0.0482	0	1	13	0.0356	0
	10000	x3	3	28	0.0867	0	1	13	0.0504	0	1	13	0.0536	0
	10000	x4	1	5	0.0231	0	1	13	0.0498	0	1	13	0.0410	0
	10000	x5	1	5	0.0224	0	1	13	0.0493	0	1	13	0.0463	0
	10000	x6	1	4	0.0233	0	1	13	0.0364	0	1	13	0.0608	0

Table 3: Test results for examples 7 – 9.

N	Pdim	SP	MKAS			PBDFF			SMDFF			SMBFGS		
			Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm
7	1000	x1	13	21	0.0255	9.49E-11	37	56	0.0302	6.92E-11	34	210	0.0942	1.64E-11
	1000	x2	11	18	0.0148	0	58	76	0.0633	9.03E-11	**	**	**	**
	1000	x3	70	123	0.1183	6.90E-11	47	71	0.0746	7.91E-11	43	251	0.0886	1.97E-11
	1000	x4	1	4	0.0096	0	65	92	0.0694	7.01E-11	**	**	**	**
	1000	x5	1	4	0.0066	0	51	70	0.0731	7.07E-11	**	**	**	**
	1000	x6	1	3	0.0073	0	54	78	0.0699	9.63E-11	15	111	0.0403	1.33E-11
	5000	x1	11	19	0.0752	8.89E-11	33	52	0.1655	4.81E-11	48	329	0.4661	3.56E-11
	5000	x2	11	18	0.0528	0	50	67	0.2263	8.72E-11	22	151	0.2341	2.96E-11
	5000	x3	11	33	0.0881	0	52	68	0.2492	9.14E-11	28	175	0.2903	6.75E-11
	5000	x4	1	4	0.0107	0	77	134	0.3275	9.66E-11	22	91	0.1943	1.43E-11
	5000	x5	1	4	0.0187	0	51	70	0.2127	1.98E-11	12	98	0.1983	3.53E-11
	5000	x6	1	3	0.0149	0	48	70	0.2395	3.51E-11	**	**	**	**
	10000	x1	9	17	0.0830	7.47E-11	33	47	0.2360	9.91E-11	33	256	0.5643	1.49E-11
	10000	x2	11	18	0.1197	0	66	87	0.4144	6.37E-11	**	**	**	**
	10000	x3	3	23	0.0781	0	44	62	0.2904	8.34E-12	19	144	0.3867	1.58E-11
	10000	x4	1	4	0.0228	0	59	87	0.3651	8.19E-11	27	110	0.3373	1.19E-11
	10000	x5	1	4	0.0225	0	56	73	0.3249	7.57E-11	12	97	0.4260	6.32E-11
	10000	x6	1	3	0.0195	0	50	65	0.3184	8.25E-11	21	74	0.2243	2.22E-16
8	1000	x1	1	4	0.0414	0	52	66	0.0825	9.67E-11	**	**	**	**
	1000	x2	18	45	0.0443	8.23E-11	71	91	0.0652	8.58E-11	36	378	0.1105	2.57E-11
	1000	x3	5	30	0.0181	0	74	93	0.1096	5.06E-11	38	380	0.2057	7.87E-12
	1000	x4	1	4	0.0123	0	79	106	0.0875	5.68E-11	12	100	0.0406	1.69E-11
	1000	x5	1	4	0.0084	0	80	97	0.1266	4.94E-11	29	349	0.1782	6.88E-11
	1000	x6	1	4	0.0126	0	57	75	0.0850	9.20E-11	19	202	0.0899	2.41E-11
	5000	x1	1	4	0.0228	0	45	68	0.2658	8.75E-11	**	**	**	**
	5000	x2	9	24	0.0855	0	57	79	0.3197	3.94E-11	35	369	0.6002	1.12E-11
	5000	x3	5	35	0.0877	0	69	98	0.3408	6.32E-11	30	313	0.5291	6.05E-11
	5000	x4	1	4	0.0192	0	63	83	0.3247	3.22E-11	19	205	0.3654	9.89E-12
	5000	x5	1	4	0.0295	0	69	85	0.3407	8.04E-11	**	**	**	**
	5000	x6	1	4	0.0301	0	63	87	0.3636	8.67E-12	14	129	0.2094	1.42E-11
	10000	x1	1	4	0.0486	0	51	69	0.4384	3.47E-11	**	**	**	**
	10000	x2	9	24	0.1131	0	73	91	0.5707	8.80E-11	45	492	1.2736	6.99E-11
	10000	x3	3	27	0.1011	0	72	96	0.5729	8.70E-11	26	261	0.8219	2.67E-11
	10000	x4	1	4	0.0238	0	73	89	0.5537	6.40E-11	16	151	0.4617	4.56E-11
	10000	x5	1	4	0.0317	0	64	87	0.5302	7.13E-11	19	229	0.6775	9.75E-12
	10000	x6	1	4	0.0250	0	51	78	0.4623	8.19E-11	15	166	0.5343	5.54E-11
9	1000	x1	1	4	0.0597	0	9	22	0.0281	7.49E-11	12	143	0.0539	2.86E-11
	1000	x2	1	4	0.0061	0	10	22	0.0300	6.63E-11	11	132	0.0317	1.35E-11
	1000	x3	1	4	0.0067	0	10	23	0.0195	1.03E-11	13	153	0.0447	3.09E-11
	1000	x4	1	4	0.0100	0	8	20	0.0133	4.82E-12	10	119	0.0370	1.27E-11
	1000	x5	1	4	0.0053	0	9	22	0.0156	5.61E-12	11	132	0.0483	2.14E-11
	1000	x6	1	4	0.0070	0	8	20	0.0153	7.83E-12	10	118	0.0521	2.22E-11
	5000	x1	1	4	0.0254	0	10	23	0.0566	3.29E-12	12	143	0.2129	6.37E-11
	5000	x2	1	4	0.0224	0	10	22	0.0605	6.86E-11	11	132	0.2053	1.37E-11
	5000	x3	1	4	0.0189	0	10	23	0.0723	2.31E-11	13	153	0.2240	6.92E-11
	5000	x4	1	4	0.0140	0	8	20	0.0603	1.08E-11	10	119	0.1744	2.85E-11
	5000	x5	1	4	0.0177	0	9	22	0.0767	1.26E-11	11	132	0.2096	4.79E-11
	5000	x6	1	4	0.0158	0	8	20	0.0464	1.75E-11	10	118	0.1745	4.96E-11
	10000	x1	1	4	0.0215	0	10	23	0.0986	4.65E-12	12	143	0.3164	9.00E-11
	10000	x2	1	4	0.0271	0	10	22	0.1266	6.89E-11	11	132	0.3047	1.37E-11
	10000	x3	1	4	0.0300	0	10	23	0.0909	3.27E-11	13	153	0.3248	9.78E-11
	10000	x4	1	4	0.0177	0	8	20	0.0820	1.52E-11	10	119	0.3020	4.03E-11
	10000	x5	1	4	0.0210	0	9	22	0.0960	1.77E-11	11	132	0.2883	6.77E-11
	10000	x6	1	4	0.0175	0	8	20	0.1014	2.48E-11	10	118	0.2954	7.02E-11